

Modeling the volumetric expansion of the lithium-sulfur battery considering charge and discharge profiles

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ABSTRACT

As an alternative battery technology to the established lithium-ion battery, the lithium-sulfur battery shows great potential due to its greater energy density, safety and possible lower material costs. In the next few years 500-600Wh kg⁻¹ could be implemented. However, there are still challenges in the commercialization and condition monitoring of the battery. Conventional methods of the lithium-ion technology, such as state of charge prediction via open circuit voltage (the discharge curve/charge curve has two plateaus), as well as coulomb counting are ineffective for the lithium-sulfur battery. However, due to the continuous volumetric change during battery operation, this could be a possible approach to monitor and predict the condition of the battery. Therefore, this paper presents a model that determines the volumetric change as a function of a constant discharge/charge profile. The influence of the selected values is investigated by a parameter variation. The parameter set for the model is further characterized by a computer tomography measurement. The simulation model is verified by literature and measurement data. This modeling approach could influence the development of a future battery management system for lithium-sulfur battery.

1. Introduction

Globally rising emissions are considered one of the main reasons for the climate change we face today. In this context, the electrification of vehicles represents an important contribution to fulfil the reduction of the global emissions. The key component of the powertrain is the energy storage system. Lithium-ion batteries (LIBs) are the first choice today but are limited in their specific energy density and safety [1–3]. A promising next generation of LIB is the lithium-sulfur (Li-S) battery. This technology shows high potential in terms of high theoretically specific energy density (2600Wh kg⁻¹), high capacity, inherent safety and abundant occurrence of low priced sulfur as active material [4,7]. However, there are still challenges in the commercialization and condition monitoring of this battery in relation to energy efficiency, self-discharge and limited lifetime (cycle and calendar aging) [4,5,8–11]. In contrast to LIB, the Li-S technology is not an intercalation process but a conversion process. In the conversion process step, the lithium ions are not deposited in a crystal structure, but a complete restructuring of the active material takes place during the lithiation and delithiation process. Inside of the Li-S battery the elemental sulfur on the

cathode side is reduced to lithium monosulfide over several phases. Furthermore, the conversion process results in a high volume change, which results in a measurable thickness change [6,12].

Due to the above-mentioned aspects of the battery with view to the complex chemistry and the challenges that still have to be overcome, it is difficult to give an accurate prediction about the current state of the battery. Therefore, there are various model approaches in the literature to describe the individual processes of the battery. The majority of models attempt to simulate the effects within the battery as good as possible. Physical, chemical and mathematical methods can be found in the literature [1–3,13–23].

All described models have their justification, but show only small parts of the individual processes within cell technology. The measurement of engineering variables, such as voltage, current, temperature will be necessary for the implementation of a Battery Management System (BMS). Conventional methods known from LIB technology, such as State of Charge (SoC) prediction using Open-Circuit Voltage (OCV), as well as coulomb counting, could be inefficiency for the Li-S battery [12,24].

The volumetric change of the cell during discharge/charge could be used as basis for the development of a future battery management

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system of the Li-S battery. The volumetric change of LIB technology has been studied many times [25–28]. On the other hand, there are only a few experimental studies of the volumetric change for Li-S technology, which investigate on pouch cell level [6]. Therefore the aim of this paper is to present a model for the Li-S battery that describes the volumetric change depending on the SoC of the battery. Thereby it is not intended to simulate various different effects of the battery as exactly as possible. The responsiveness of the model approach is investigated by a sensitivity analysis. Based on the results of the analysis, the model approach is validated with measurements from literature and experimental measurements on prototype pouch cell at different C-rates. The parameterization of the model for the prototype pouch cells is performed using computer tomography (CT) scans. Thus, it will be shown that the model can determine the volumetric change as a function of the SoC.

2. Methodology

In the Li-S cell, various material conversions take place in which solids are dissolved or precipitate from the solution. The applied theoretical methodology results from the conversion behaviour of the cell during its operation. In order to be able to describe the resulting volume changes, these reactions will be considered in more detail.

2.1. Structure of the paper

The work flow in Fig. 1 describes the approach in this paper. At the beginning, the mechanical behaviour of the Li-S cell will be modelled. Within the Li-S cell, various complex chemical process steps take place, so that the modelling is carried out on different levels. For both electrodes, the elementary processes at the molecular level are considered first. This includes a quantification of the converted active material as well as the volumes that are released or taken up by it. Based on this, a transfer to the electrode level is carried out. This transfer is not trivial, as for example the porosity of the cathode has a great influence. Finally, the results are combined to describe the change in thickness of a full cell.

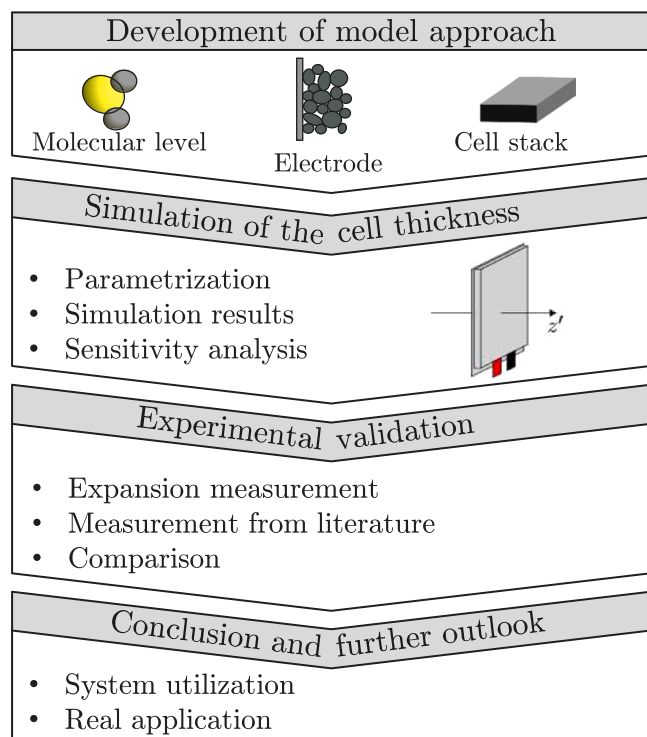


Fig. 1. Workflow scheme used for the model approach and the derivation and validation.

The model is created in Matlab/SIMULINK to simulate the volumetric change. The influence of the parameter set used for the model is investigated in a sensitivity analysis. The simulation results are validated with experimental investigations on a Li-S test cell and from the literature. Aging effects on the battery are partially considered in this model and are only taken into account with a factor for the anode side. The irreversible expansion on the anode side is described in more detail in [17,23,29].

2.2. Volumetric changes of the electrode within Li-S cells

During the discharge of the battery, elemental sulfur S_8 is reduced on the cathode surface to soluble intermediate stages, the polysulfide chains. As the discharge process continues, the polysulfide chains become shorter until they precipitate as Li_2S_2 as solid form at the cathode, where they continue to shorten until Li_2S . At the end of discharge, the majority of the polysulfides on the cathode side should be Li_2S . This depends on the operating conditions (voltage limits) [6,9]. Hence, four transition stages occur during the discharging and charging process in the battery: solid to liquid, liquid to liquid, liquid to solid and finally solid to solid. Due to the conversion process within the battery, the density state between the starting product elemental sulfur (S_8) and the end product lithium sulfide (Li_2S) also changes. Sulfur (S_8) has a density of 2.03 g cm^{-3} and Li_2S a density of 1.67 g cm^{-3} . In this process each mole of sulfur is converted into eight moles of Li_2S , which implies that the cathode must provide an extra volume of 80% [6,30]. Due to the lower density of lithium sulfide than the elementary form of sulfur, it can be deposited much deeper into the pores. Experimental studies have shown that the cathode is 22% thicker in the discharge state than at the beginning [31]. Furthermore, it has been shown that the morphology of Li_2S is dependent on the current rate [32]. Fan et al. show that at low current rates larger crystals are formed, whereas at higher discharge rates rather fine structures are observed [6,32]. The charging process takes place in reverse order so that Li_2S oxidises and longer polysulfide chains are formed until elemental sulfur (S_8) precipitates again on the cathode surface.

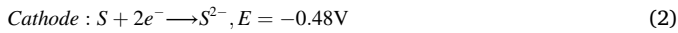
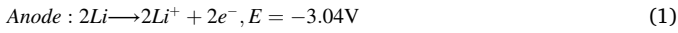
In the process stage of dissolved polysulfides, the thickness of the cathode is at a minimum and increases again as the sulfur precipitates. Experimental studies have shown that the thickness is 4% greater than in the discharge stage [31]. With regard to the total cell expansion, the volumetric change of the lithium anode dominates, which masks the cathode expansion. With respect to the total cell expansion, the volume change of the lithium anode dominates and the expansion of the cathode gives the curve shape its typical look. This behavior can be seen in the simulation results of the model in chapter 5. Thus, the battery contracts during discharge and reaches its maximum thickness during charging [6].

The capacity decrease as well as the side reaction based on the shuttle mechanism is not considered in this model approach. The mechanism is based on dissolved polysulfide chains. These polysulfide chains are no longer converted back into elemental sulfur during the charging process. These remaining higher order polysulfide chains can pass through the separator to the anode. At the metallic lithium anode, a parasitic reaction would occur with the higher order chains. This reaction can form lower order polysulfides, which can diffuse back to the cathode and recover higher order polysulfide chains [12,33–36]. The transport of the sulfur species to the anode and the subsequent chemical reduction at the anode surface. During this process, reversible and irreversible capacity decreases occur [36]. By using lithium nitrate ($LiNO_3$) as an additive in the electrolyte, a stable Solid Electrolyte Interface (SEI) can be formed on the anode [37]. This reduces the parasitic reaction and can thus suppress the shuttle mechanism [12,34]. In addition, a high sulfur/electrolyte ratio could inhibit species transport and hinder the shuttle mechanism. In contrast, this would have a performance-decreasing effect on the whole cell [36,38]. Thus, the shuttle mechanism will be excluded from the presented model because

LiNO₃ is used in the test cells.

3. Model

The Li-S cell contains two electrodes, the anode which consists of pure lithium metal and the cathode of a sulfur/carbon mixture. These two electrodes are covered by an organic electrolyte and separated from each other by a separator, which is an electrical insulator but lithium-ion permeable. The reaction that takes place can be described for the discharge process by the following partial equations:



Since sulfur on the cathode side is present in elementary form (S₈ molecule), the total reaction results:



The battery voltage in the charged state oscillates between 2.1–2.4V. The voltage curve as a function of the SoC is shown in Fig. 2 at a constant discharge/charge current. The displayed profile shows that the voltage does not drop continuously but presents two plateaus with a low point. The two regions can be described as upper and lower discharge plateaus. In addition the lower discharge plateau is almost constant over a large area before the curve drops abruptly. The charging curve is not constant either and shows increases at the end as well as at the beginning, which can be attributed to the conversion from solid to solid. The characteristic development of the discharge or charge curve can be attributed to the complex chemical conversion stages. Lin and Qi describe the origin of the plateaus and relate them to the dissolution/precipitation of the polysulfide chains, which can be prevented by a highly concentrated electrolyte or by minimising the pore size of the carbon/sulfur compound [19]. Due to these factors, the course of the voltage curve can change and a precise assignment to the SoC of the battery becomes more complicated. Waluś et al. [6] show that there is a correlation between the capacitance and the variation in cell thickness.

3.1. Modelling of the anode

For modelling the anode it is necessary to know the amount of required lithium, which must be converted. This amount depends on the sulfur concentration of the cathode. In theory, the anode would consist of the same amount of metallic lithium so that the total elemental sulfur can be transformed. In practice, more lithium is required than the theoretical amount, as side effects still occur. Due to a passivation layer, the SEI on the anode side, the shuttle effect should be suppressed, but this has a negative influence on the efficiency of the cell [8,14,18,39].

For the model of the anode, a behavior proportional to the charged/

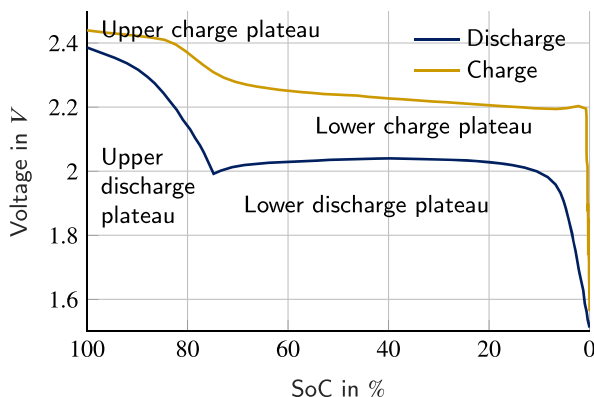


Fig. 2. Voltage curve for a discharge and charge process.

discharged capacitance is assumed. This assumption can be made because the electrochemical reaction of oxidation takes place during the discharge/charge process with a constant current. To simplify the model approach, homogeneous and linear deposition is assumed. By cycling the cell without further countermeasures, dendrite growth would occur on the anode side. This would not be homogeneous and linear. In several studies it is shown that dendrite growth can be inhibited by the influence of an external pressure [40–43]. Barai et al. research show the impact of an external pressure on dendrite growth [40]. It has been found that there is a range of conditions of critical pressures at which the growth of dendritic lithium protrusions can be prevented. Plastic deformation of lithium at high pressure results in reduced dendrite growth and more stable deposition of the metallic lithium. The test cells would be strained below 3.14kg cm⁻² in regular operation to cause more stable deposition and achieve higher cycling. This process is shown in Fig. 3(a). Lithium atoms have one valence electron. During the discharge process, for every electron that has reached the cathode side via the outer circuit, a lithium ion also diffuses through the electrolyte to the cathode side. This process can be described by Faraday's law, which defines the relationship between electric charge and mass transfer. Thus, it is possible to calculate the amount of converted lithium n_{Li} based on the relationship of Faraday's laws by knowing the amount of charge quantity Q that has flowed per cell:

$$n_{\text{Li}} = n_e = \frac{Q}{q_e \cdot N_a} \quad (4)$$

In this context, q_e is the elementary charge and N_a is the Avogadro constant. The amount of electrical charge can be calculated as a function of the nominal capacity C_N of the cell and the current SoC (SoC_0) in percent. To scale the amount of electrical charge that has flowed on one anode layer, the nominal capacity of the cell is divided by the number of active areas N_{Stack} :

$$Q = \frac{C_N}{N_{\text{Stack}}} \cdot (100\% - \text{SoC}_0) \quad (5)$$

The volume of the converted lithium can be calculated with Eq. 4 and

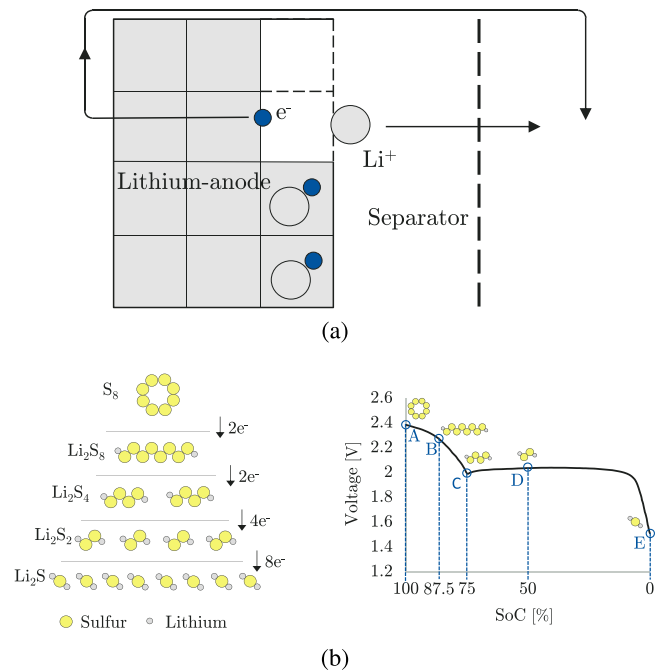


Fig. 3. The schematic volume change of the anode during discharge is shown in (a). The reduction of the polysulfide chains and the corresponding discharge voltage curve are illustrated in (b).

the molar volume of lithium $V_{m, Li}$. For the calculation of the molar volume, the density ρ_{Li} and the molar mass M_{Li} of the cell must be known.

$$\Delta V_{Li} = n_{Li} \cdot V_{m, Li} = n_{Li} \cdot \frac{M_{Li}}{\rho_{Li}} \quad (6)$$

Due to the reason that the anode consists almost totally of metallic lithium without porosity, the volume change of lithium also corresponds to the volume change of the whole anode.

Experiments by Rieger show that the elongation of the electrode in x- and y-direction is small compared to the elongation in z-direction [44]. For this reason, an elongation of the anode in z-direction only is assumed. A further assumption for a simplified modelling of the anode is a homogeneous distribution of the thickness change over the electrode surface (no gradient of the z-elongation in x- or y-direction). This assumption is supported by the homogeneous structure of the anode. With the cross-sectional area A of the cell vertical to the z-axis, it is possible to calculate the thickness change from the volume change. Based on the described assumptions and Eqs. (4)–(6), the relationship for the change in thickness for an individual anode can be established as follows.

$$\Delta t_{An} = \frac{\Delta V_{An}}{A} = \frac{\Delta V_{Li}}{A} \quad (7)$$

$$\Delta t_{An} = \frac{C \cdot (100\% - SoC_0) \cdot M_{Li}}{q_e \cdot N_A \cdot \rho_{Li} \cdot A \cdot N_{Stack}} \quad (8)$$

Thus the modelled anode is represented by a cuboid with one degree of freedom, which is the length in z-direction. A reversible behaviour is assumed for the charging process.

3.2. Modelling of the cathode

The cathode of the Li-S battery does not consist only of the active material sulfur, as this is not electrical conductive. To enable charge and particle transport from the current collector, the sulfur is embedded in a porous carbon structure. The amount of sulfur in the total mass of the dry cathode is usually 60% to 85%. To benefit from the gravimetric energy density of sulfur compared to the active materials used in lithium-ion technology, the mass fraction should be at least 70% [45]. In the sulfur cathode, different pore sizes occur. These are important for modelling the expansion depending on the SoC. While in the macro- and mesopores the major part of the mass transport or storage takes place, in the micropores mainly the end products (Li_2S_2 and Li_2S) of the reaction are stored [12]. Typical values for the porosity of the sulfur-carbon composites are 50% to 80% [45]. In the Li-S battery, the lithium ions react with the elemental sulfur on the cathode side and finally deposit as lithium sulfide (Li_2S) via various intermediate stages (lithium-polysulfides). These individual intermediate stages are shown in Fig. 3b. In this paper, for the description of the volume change at the molecular level, it is assumed that the individual partial reactions occur sequentially. Since in reality the processes run simultaneously, this will be taken into account in a later calculation. If other intermediate stages occur, these are only present in dissolved form and can therefore be neglected when calculating the solids-related volume change.

In order to completely reduce an S_8 molecule to Li_2S , 16 lithium atoms and 16 electrons are needed. Thus, it is possible to assign the respective SoC by the number of electrons that have been required (Fig. 3b). In the initial state (point A), solid sulfur is present in the form of S_8 molecules (crown-shaped ring). 16 absorbed electrons correspond to the complete reduction, also a SoC of 0% (point E), and zero absorbed electrons correspond to the starting point A of discharge, also a SoC of 100%. At point B it is assumed that the highest concentration of Li_2S_8 exists in the electrolyte. At this point all solid sulfur is dissolved in the electrolyte and therefore does not take up any solid volume in the

cathode matrix. At point C, there is the transition to the lower discharge plateau. Therefore Li_2S_8 is reduced to Li_2S_4 . The reduction takes place in dissolved form. At point D (SoC = 50%) Li_2S_2 is a solid which influences the volume of the cathode. The conversion of Li_2S_2 into Li_2S is recrystallization process from one solid to another. In the most cases a mixture of Li_2S_2 and Li_2S is present at the end of the discharge, because the discharge process is terminated when the voltage drops below the end of discharge voltage [12]. Therefore it is assumed that there is a quantitative ratio of the two end products depending on the discharge rate (C-rate). The higher the C-rate, the less time is left for the slow solid conversion and thus the proportion of Li_2S in the total product decreases. This proportionality has not yet been investigated in detail and therefore a functional relationship is assumed. The relationship between the two solid end products Li_2S_2 and Li_2S is described by the following equation:

$$ratio_{LiS} = \frac{n_{Li_2S}}{n_{Li_2S} + n_{Li_2S_2}} \quad (9)$$

Furthermore, the coulomb efficiency is determined by the ratio of the removed capacity (C_{rem}) to the charged capacity (C_0) equals the difference between one and the Depth of Discharge:

$$\eta_{Ah} = \frac{C_{rem}}{C_0} = 1 - DoD \quad (10)$$

It is assumed that in case of whole discharge of the battery ($\eta_{Ah} = 1$), the complete Li_2S_2 is also converted into Li_2S ($ratio_{LiS} = 1$). If the battery is half discharged ($\eta_{Ah} = 0.5$), no Li_2S (see Fig. 3 (b)) has been formed ($ratio_{LiS} = 0$). Thus, a linear equation can be determined as a function between the $ratio_{LiS}$ and C_{rem} :

$$ratio_{LiS} = 2 \cdot \eta_{Ah} - 1 = 2 \cdot \frac{C_{rem}}{C_0} - 1 \quad (11)$$

The Eq. 11 applies to small C-rates. The assumption for the sequential conversion of sulfur be true to infinitesimally small currents. Since the Li-S cells do not have such high power capability at the pouch cell level, this equation does not apply at a C-rate of more than 0.3C and will be approximated with a decreasing exponential curve. The volumes of active material in the points shown in Fig. 3(b)) can be determined from the density and the molar mass. Table 1 shows the calculation for the conversion of one mol sulfur. The solved intermediate stages in the phase transformation are set to zero, because they are not considered in the calculation of the cathode expansion.

Due to the structure of the cathode consisting of carbon and elementary sulfur and its porosity, the volume changes influence the whole expansion of the cathode only to a limited extent. Thus, there exists a correlation between the porosity and the expansion of the cathode. The dissolution and precipitation of the active material causes local pressure differences in the cathode structure. These are compensated by expansion/contraction of the entire carbon grid or by the change in pore volume due to electrolytes flowing in and out. Theoretically, the pore volume should be able to absorb the increase in volume of the active material of 80%. By this assumption, the cathode volume would be constant over the duration of the discharge. This is not the

Table 1
Volumetric change of the active material in the cathode.

Point	A	B/C	D	E
Intermediate	S_8	Li_2S_8/Li_2S_4	Li_2S_2	Li_2S
Density [$g\ cm^{-3}$]	2.07		1.80	1.66
Molar Mass [$g\ mol^{-1}$]	256.5		78	46
Molar Volume [$cm^3\ mol^{-1}$]	123.9		43.3	27.7
Amount of Substance [mol]	1		4	8
Volume [cm^3]	123.9	0	173.2	221.6
Volume Change [%] (related to point A)	-	- 100	+ 39.8	+ 78.9

case, because the change in volume of the active material results in a change in the pore volume as well as in an expansion of the cathode. The description by Barai et al. of the pore mechanical effect in the cathode of the Li-S battery assumes that the cathode in point B of Fig. 3b contracts only slightly (approximated 4%) [16]. The carbon framework in combination with the binder remains almost form stable up to this point and the released volume is filled up by the electrolyte. In points D and E of Fig. 3b, a constant pore volume is assumed. This results in a calculated cathode elongation in z-direction at the end of discharge in the range of 15–25%, which corresponds well with measurements by He et al. of about 22%. In summary, it is assumed that the pressure compensation at a negative pressure (at point B) is mainly achieved by changing the pore volume. In the case of an over-pressure (point D and point E), the pressure compensation is mainly due to an expansion of the cathode structure. This assumption can be assumed because the capillary effect makes it easier for the electrolyte to flow into and out of the cathode. In addition, the modulus of elasticity of the carbon framework varies depending on the load by tension or pressure. This means that the cathode behaves differently under negative pressure than under positive pressure [16,31]. In addition, the infiltration of the Li_2S particles into the micropores expands the cathode framework. In this way the elongation can be determined for the four points (A, B, D and E). The curve of the cathode expansion is performed by using a spline interpolation between the four points, because this corresponds most closely to the real case. The expansion at the interpolation points depends strongly on the composition of the cathode. The higher the sulfur content compared to the carbon in the cell, the greater the change in elongation. Brückner et al. investigations have shown that above a mass fraction of 80% there is no significant sulfur utilization [46]. To describe this connection a factor y_{cond} is introduced, which depends on the mass fraction of the sulfur. For this purpose a lookup table is used, which was developed based on Brückner et al. investigations. This factor y_{cond} is multiplied by the amount of sulfur to calculate the amount of sulfur involved in the reaction. The sulfur that is not available is included in the calculation with a constant volume. In the non-idealised case, the chemical processes taking place in the cathode will run simultaneously. For this purpose, the curve of the cathode thickness is smoothed in the following equation:

$$\Delta t_{\text{Cat}}(t) = \alpha \cdot t_{\text{Cat,spline}}(t) + (1 - \alpha) \cdot t_{\text{Cat,linear}}(t), \quad 0 \leq \alpha \leq 1 \quad (12)$$

The curve is composed of the interpolated part and a linear part. The linear equation for the linear part is determined with the elongations at the beginning and end of the discharge. The determining factor for the weighting of the two parts is the linearity factor α . This value can be set separately for the discharge and for the charge. Thus a hysteresis can be implemented if the factors for charge and discharge differ. The factor is also dependent on the C-rate. At low discharge rates it can be assumed that the resulting Li_2S particles are comparatively large and irregular, whereas at high C-rates they are small and uniform.

3.3. Model implementation

The models for the volumetric change of the anode and the cathode are implemented in MATLAB/Simulink. Fig. 4 shows the schematic model structure. This structure is divided into the upper part for the determination of the anode thickness and the lower part for the determination of the cathode thickness. The areas with a blue highlight represent constant values. These include the thickness of the separator, the current collector and the initial thicknesses of the anode and cathode. The grey and yellow blocks represent the input parameters. Here, the yellow blocks are variable values and the grey blocks are parametric values. The five input parameters include the number of discharge/charge cycles, the C-rate in form of a current profile, the sulfur loading in mg cm^{-2} , the carbon- content relative to the mass of the dry cathode (without electrolyte) and the porosity of the cathode. The output signal of the model can be the battery thickness. With the C-rate and the calculated effective capacity C_{eff} , the edge amplitude I , as well as the period T_{cyc} of the square wave signal are determined according to the following equations:

$$I = C_{\text{eff}} \cdot C\text{-rate} \quad (13)$$

$$T_{\text{cyc}} = \frac{3600s}{C\text{-rate}} \quad (14)$$

In this way, the battery is always fully discharged and then charged with a constant current. In this context, fully does not necessarily mean up to a SoC of zero as explained in Section 3. For the calculation of the SoC, the maximum capacity is used as reference:

$$\text{SoC}(t) = 1 + \frac{1}{C_{\text{max}}} \int_0^t I(t) dt \quad (15)$$

The maximum capacity can be calculated with the amount of substance of sulfur according to Eq. 16, since a maximum of 16 electrons can be used for each sulfur molecule.

$$C_{\text{max}} = n_s \cdot N_A \cdot 16 \cdot q_e \cdot N_{\text{stack}} \quad (16)$$

The sulfur loading load_s can be determined from the cross-sectional area A of the cell and the amount of sulfur present.

$$n_s = \frac{A \cdot \text{load}_s}{M_s} \quad (17)$$

Thus, from the composition of the cathode and the determination of the battery values, it is possible to determine the interpolation points for the expansion curve. Based on these values, the current cathode thickness can be determined. After determining the individual electrode thicknesses, these and the two constant thicknesses of the collector and the separator are summed together. By knowing which number of layers the cell consists of, the thickness of the cell is calculated. Subsequently, this is multiplied by the number of stacked cells to estimate the whole bat-

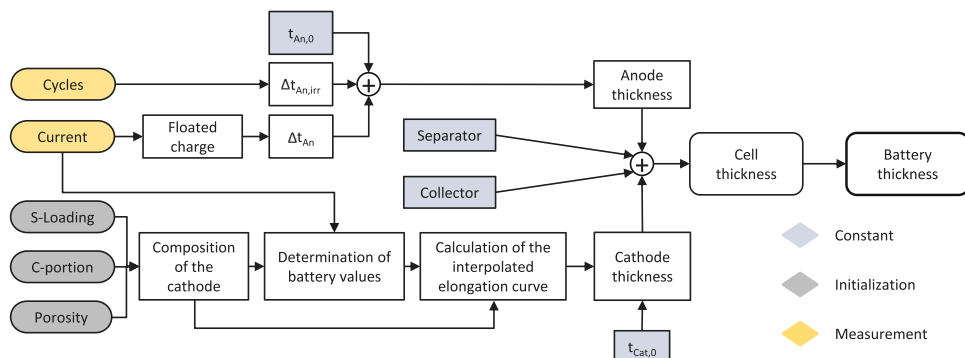


Fig. 4. Workflow scheme used for the model approach and the derivation and validation.

tery thickness. This approach to determining the total thickness of the cell is intended as a first approximation. In this modelling approach, the extension of one cathode and anode layer is determined mathematically and then scaled up to the total cell. This enables a faster calculation time. Nevertheless, it cannot be excluded that a cooperative error occurs with regard to the total thickness of the cell/battery.

4. Experimental

The developed model is validated by measurements on pre-commercial Li-S pouch cells. The tested pouch cells are manufactured by the Fraunhofer Institute for Material and Beam Technology IWS [5]. The Table 2 summarizes the main parameters of the pouch cells.

The following measurements are divided into two parts. To parameterize the model, a CT measurement is performed. In our case, the CT measurement should identify the individual thicknesses of the battery's components, such as the anode, cathode, separator and current collector.

To validate the model approach and simulation results, the volumetric change of the cell at different C-rates is investigated. For this purpose, thickness measurement at different discharge C-rates are performed. At the beginning of the measurement, an initial measurement is done to determine the real capacitance.

4.1. Computer tomography

At BoL, a CT measurement of the Li-S test cell is performed. Based on the CT scans, the general structure of the cell can be seen without it having been cycled. The CT scans are taken by an industrial CT scanner of the company Zeiss (METROTOM 1500). A 3D data set with absorption levels is reconstructed from the X-ray scans. The individual components of the battery cell can be clearly identified due to their different densities. Materials with a high density appear brighter in the scans than materials with a lower density.

Fig. 5 shows different CT scans at two different resolutions. Fig. 5(a), (b), and (c) were taken at a resolution of $0.049\mu\text{m}$ per voxel and Fig. 5(d), (e), and (f) at a resolution of $0.016\mu\text{m}$ per voxel. Fig. 5(a) shows the whole test object. It is visible, that the Li-S cell is a pouch cell in which the electrode stack is encased in an aluminum laminate. Fig. 5(b) shows a cross-sectional view from above of an internal surface of an electrode. This allows the determination of the dimensions of the cathode/anode surface. The width of the electrode stacks is 46mm and the height is 70mm . Based on the measured height and width, an area of 32.2 cm^2 can be calculated. In addition, both current collectors are visible. In Fig. 5(c) is a cross-sectional view from above through the cell. In this sectional view, the individual double-sided cathodes are identifiable. The current collector on the anode side is clearly detectable, as it is made of copper. It can also be observed that this is a cell, as the individual cathode layers have different widths. The thickness of the whole cell can be determined to 10mm . Furthermore, the number of cathodes and anodes are necessary for the model. The number of layers can be identified in Fig. 5(d) on 19 double-sided cathodes and on 18 double-sided anodes and 2 single-sided anodes (at the edge of the pouch cell). The porosity of the cell is about 60%. The individual pores are soaked with electrolyte and are rather dark in the CT scan. The thickness of a double-sided cathode can

be determined to be 0.33mm to 0.44mm . In Fig. 5(e), the polyethylene (PE) separator ($12\mu\text{m}$) and the lithium anode ($2 \times 50\mu\text{m}$) are found between two double-sided cathodes. These two components are not very well visible on the CT scans, due to their small dimensions and low density. In the same sub-Fig. 5(e) it is visible that the copper current collector has not the same length throughout the anode. In the upper part, the collector has a length of approximately 11mm . This serves the mechanical connection to a load. In the further course, the current collector is designed thin, which is nearly invisible. In the case that the collector had the same thickness throughout, it would have a significant negative impact on the gravimetric energy density. In the last Fig. 5(f), a cut through the cell is displayed in a zoomed view. The porous structure of the cathode and the shortened current conductor are clearly visible here. It can be also seen that the individual surfaces are slightly corrugated in the cell.

4.2. Volumetric measurement

The performance of the model approach is tested and analyzed by different measurements on our cell. The measurements can be divided into two parts. There is an initialization measurement and expansion measurements at different C-rates. The measurements are carried out once on the cell. The measured cell type is subject to a fast capacity fading. The capacity fading can be attributed to gas formation by lithium nitrate in the electrolyte on the one hand and to the deposition of insoluble and irreversible sulfide species on the other hand [6,47–49]. The measurement is performed with a test setup from Digatron (MCT-10-06-12 ME). The resolution of the test circuit is ± 15 bit and corresponds to an accuracy of $\pm 0.1\%$. The Li-S cell is measured at a temperature of 25°C in a climatic chamber. A four-point measurement is taken on the test cell. The extension is recorded via a dial gauge. The measured values of the measuring sensors are sampled by the NI 9237 bridge module. The following procedure is carried out for the Li-S cell:

1. Tempering the cell at 25°C for 2h
2. Initial cycle to determine the cell capacity
 - (a) Discharge with 0.2 C-rate (1A) until the minimum voltage (1.8V) is reached. This step resets the history of the battery
 - (b) Relaxation the battery cell for 2 hours at the test temperature
 - (c) Full charging of the battery cell. CC charge with 0.1 C-rate (0.5A) until the end-of-charge voltage (2.6V) is reached
 - (d) Relaxation of the cell for 2 hours at the test temperature
3. Relaxation and tempering the battery cell for 1 hours at the test temperature (25°C)
4. Modification of the test setup for the expansion measurement (Connect the dial gauge)
5. The following sequence represents the expansion measurement and is repeated for the C-rates of 0.05, 0.1, 0.2 and 0.5
 - (a) Recording of the expansion at full charge (SoC = 100%)
 - (b) Discharge with the first C-rate until the minimum voltage (1.8V) is reached
 - (c) Thereby a continuous recording of the expansion of the cell
 - (d) Relaxation of the cell for 2 hours, with no current passing through the cell
 - (e) Charge with 0.1 C-rate (0.5A) until the maximum voltage (2.6V) is reached
 - (f) Thereby a continuous recording of the expansion of the cell
 - (g) Relaxation of the cell for 2 hours, with no current passing through the cell

During the initialization measurement, 2.5Ah could be charged into the cell. The next part of the measurement involved the expansion of the test cell. The expansion measurement is recorded in millimeter by using the dial gauge. Fig. 6(a) shows the different voltage curves at the different C-rates plotted against time. It can be seen that the higher the C-rate, the less charge can be taken from the cell. This can be seen in the reduction

Table 2
The main parameters of the test pouch Li-S batteries .

Parameter	Value
Nominal Capacity	5Ah
Nominal Voltage	2.1V
Maximum Voltage	2.6V
Minimum Voltage	1.8V
Nominal Charging Current	0.5A (C/10 rate)
Nominal Discharging Current	0.25A (C/20 rate)
Temperature Operation Range	-10 to 40°C

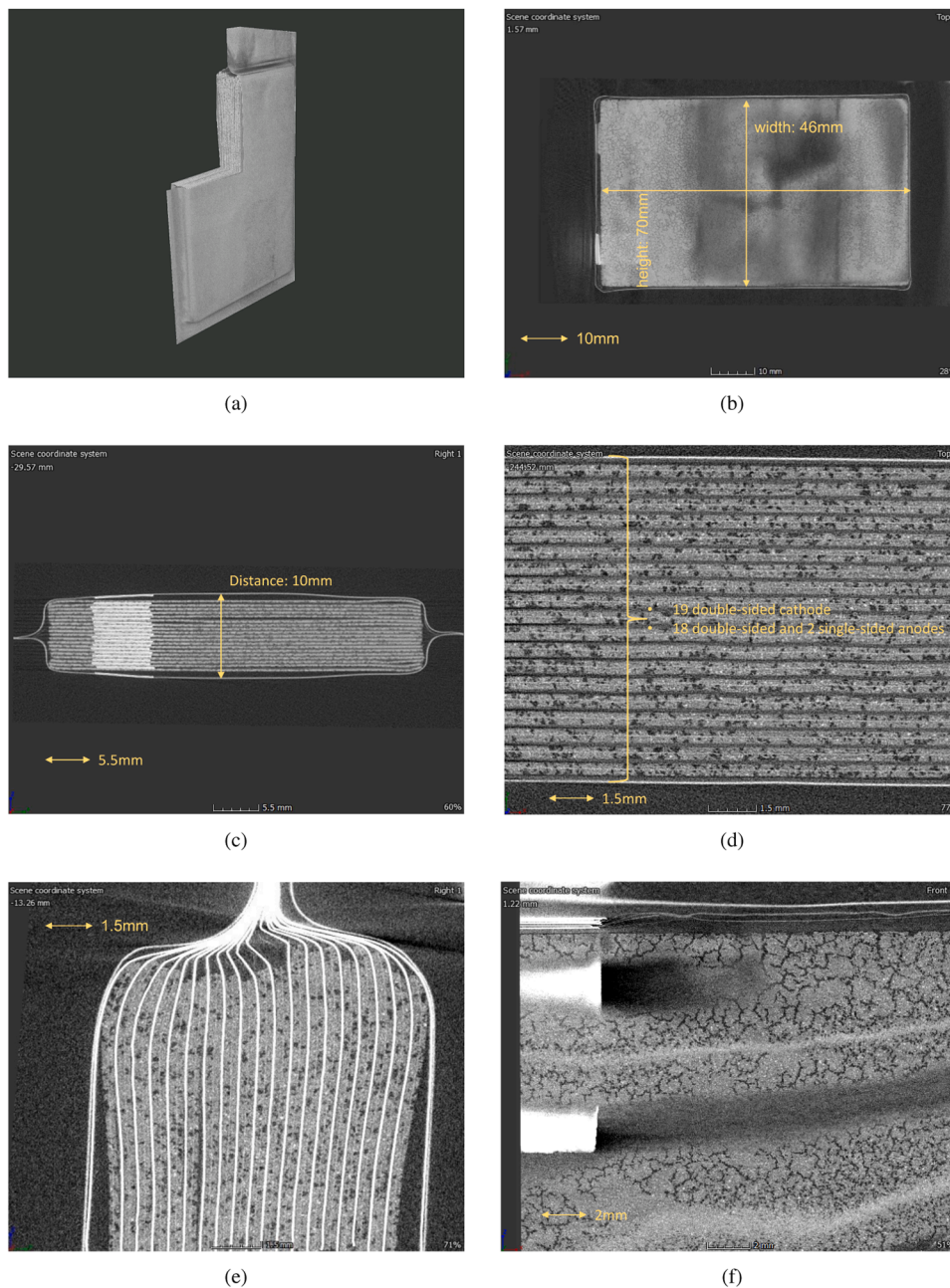


Fig. 5. CT scans of the Li-S cell at different resolutions. Images (a), (b), and (c) are taken at a resolution of $0.049\mu\text{m}$. Image (a) shows the whole cell. A single cathode layer from above is shown in (b). The thickness of the whole cell and the individual layers can be seen in (c). Fig. (d), (e) and (f) are taken at a higher resolution of $0.016\mu\text{m}$. In (d) the double-sided cathodes are shown. In (e) is a side cut through the cell in which the current collector is visible. Image (d) shows a porous cathode layer from above.

of the voltage intervals. The yellow highlighted areas, show the voltage ranges at the different defined C-rates. It can also be observed that in the case of a previous discharge with a low C-rate, the charge curve has a noticeable overvoltage at the start of the charge. These overvoltages decrease with increasing C-rate. In this pause steps the voltage increases to 2.1V.

The corresponding thickness changes of the voltage profiles can be seen in Fig. 6(b). As in Fig. 6(a), the areas marked in yellow are the volumetric change at the different C-rates. It can be seen that the cell contracts during discharge and expands during charge. In addition, the cell thickness decreases in nearly all pause steps. The thickness changes show a very linear characteristic in the discharging direction and a spline characteristic in the charging direction. In dependency of the C-rate The higher the discharge C-rate, the less the cell contracts. In the charging direction, the cell expansion increases a little per cycle.

5. Results and discussion

The experimental investigations show that different C-rates result in volumetric formation at the cell. At first, the measurement results are classified in the general context of the model. Subsequently, the implemented model approach is simulated in MATLAB/Simulink. The influence of the selected parameters is investigated by parameter variation. The expansion measurement serve to validate the model concept.

5.1. Expansion measurement at different discharge rates

At first the results of the expansion measurement are used to validate the theory of the implemented model approach. As described in Section 3, various conversion processes of the polysulfide chains take place within the Li-S cell during the discharge/charge process. The conversion and deposition of the individual polysulfide chains are accompanied by a change in density and thus a change in volume. These can be observed

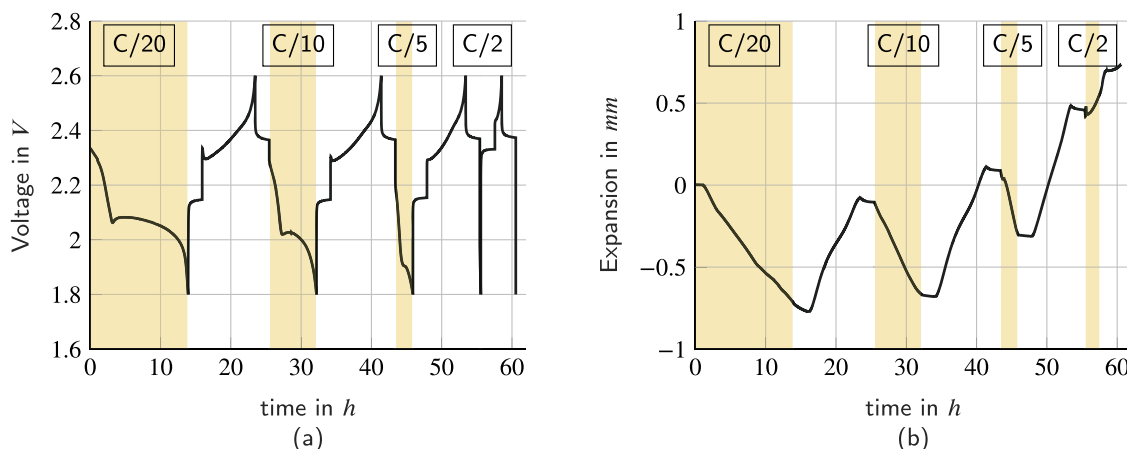


Fig. 6. Fig. (a) shows the different discharge profiles over time and the corresponding thickness changes (b) over time of a 5A cell. In the yellow highlighted area, the cell is loaded at the defined different C-rates. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in the curve of the expansion measurement and the associated voltage curve in Fig. 6(a) and Fig. 6(b).

Fig. 6(a) shows the different voltage curves at the different discharge C-rates. At the C/20 discharge C-rate, the two described voltage plateaus in Section 3 are best seen. As soon as the voltage curve reaches the low point, it is assumed that the precipitation of the solid Li_2S_2 begins. During this time, the elemental sulfur has been dissolved in the electrolyte and reduced once. Thus, the density of the polysulfide chain has been reduced. The precipitation of the solid Li_2S_2 slightly inhibits the change of the expansion, cause at the same time the solid is deposited in the micropores on the cathode side. This process step is in Fig. 6 (b) unrecognizable in the discharge directions, but in the subsequent charge direction. A small turning point is visible in the expansion curve. The rapid voltage drop at the end of a discharge pulse indicates the final conversion of the Li_2S_2 to Li_2S . This results in an ion-impermeable passivation layer on the cathode side, which increases the internal resistance.

In the charging steps in Fig. 6(b), the conversion processes of the polysulfide chains are also evident through the change of expansion. At the beginning of the charging steps, there is a voltage peak. In this process step, the solid Li_2S_2 and Li_2S is dissolved, which requires a higher energy. The higher the discharge C-rate before a charging step, the lower this voltage pulse is. This indicates a lower deposition of the solid Li_2S_2 and Li_2S in the previous discharge step. The higher the discharge C-rate, the smaller the cell expansion in Fig. 6(b). This indicates a lower utilization of sulfur. It is visible that the expansion curves of the charge pulses become more linear. This indicates the assumed moss-like structure of the solid Li_2S_2 and Li_2S due to higher discharge C-rates. This structure can be dissolved faster than large crystals. The larger the crystals of the solid from the previous discharge pulse, the slower the dissolution process, which is reflected in a small turning point of the charge curve. As the solid sulfur (S_8) is precipitated at the end of a charge step into the macropores of the graphite matrix, the voltage curves in Fig. 6(a) drop is fast increasing. The course of the voltage ends at the formation of an ion impermeable surface layer. Finally, the curve of voltage and the expansion change in discharge/charge direction can be used to describe the modeling of the cathode and anode processes and the individually assumed polysulfide stages.

Fig. 6(b) shows that the expansion curve has increased in volume after several cycles. As already mentioned, LiNO_3 is added to the electrolyte as an additive so that a protective layer can form on the anode. The reaction of LiNO_3 with the metallic anode can also result in the evolution of gases, which can cause irreversible swelling of the pouch cell. In addition, the cell is not strained during its thickness measurement, so inhomogeneous deposition of the lithium cannot be ruled out.

In [6], a post mortem analysis of a single layer Li-S cell has been performed and related to the reactions within a pouch cell. The results show that the response of the global cell is mainly determined by the behavior of the metallic lithium anode, which masks the volumetric change of the cathode. The characteristic curved shape is influenced by the reactions of the cathode.

5.2. Parameter study and simulation results

In order to validate the implemented model approach, expansion measurements have been performed at different C-rates. Due to the small number of measurement curves and the not known exact composition of the cell, an additional parameter variation is performed to investigate the sensitivity of the implemented model. The parameters and variables which are used for the simulation are listed in Table 3. Values marked with (*) represent model parameters which are calculated by the assumed material composition. The geometric dimensions of the cell components listed in the table are derived from the CT scans. The required mathematical relationships are based on the publication of Dörfler et al. [5]. The simulation results of the thickness change for one cell and the corresponding parameter variations are shown in Fig. 7. The sensitivity analysis is performed for the following four variables: C-rate, sulfur loading, porosity and graphite content.

The thickness profiles of the cell are shown in all the diagrams in Fig. 7. Thereby, the waveform represents the volume change of the anode and the cathode. The spline course is visible in all sub-figures. This waveform can be identified in the experimental investigations (see Fig. 6(b)) during the charging process. The cell contracts during discharge and expands during charging. The variation of the model parameters has an impact on the maximum expansion, the removed capacity and the initial thickness of the cell. A total thickness of 10mm has been determined from the CT images. Depending on the choice of parameters, the calculated initial thickness will vary up to 25%. The deviations from the model can be explained by the not achieved series production readiness of the cells. From the CT scans, it can be seen that about 20% of the total thickness of the cell assumes voids.

Fig. 7(a) shows the expansion curves for three different C-rates versus the accumulated charge amount. The comparison of the curves with each other illustrates the increasing linearity with rising C-rate. It is also noticeable that at a high C-rate, the amount of capacity that can be removed from or added to the cell decreases. Therefore, the modification of the C-rate has no impact on the initial thickness, but on the course and the achievable expansion/contraction of the cell.

The sulfur loading represents an important issue in the achievable specific energy density. Therefore, the composition of the cathode is

Table 3

Model parameter.

Parameter	Description	Unit	Value
Physical constants			
F	Faradays constant	C	9.649e4
M	Molar mass S	g mol^{-1}	32
N_A	Avogadro constant	mol^{-1}	6.0221e23
z	Electron number per reaction related to one molecule	-	2
$n_{S_0}, n_{S_4}, n_{S_2}, n_S$	Number of S atoms in polysulfide	-	8, 4, 2, 1
q_e	Elementary charge	C	1.602176634e-19
Cell design properties			
x_{Bat}	Width of the battery	mm	50
y_{Bat}	Length of the battery	mm	70
A_{Bat}^*	Area of one cathode/anode/current collector/cell stack	cm^2	35
t_{Bat}	Initial thickness of the battery	mm	10
$t_{An,1}$	Initial thickness of the anode	mm	50
t_{Sep}	Thickness of the separator	μm	12
t_{Coll}	Thickness of the aluminium collector	μm	20
n_{St}	Number of layers	-	38
Operational parameters			
C_{max}^*	Maximal capacity	Ah	~ 5.0
C_{eff}^*	Effective capacity	Ah	~ 3.2
ρ_S	Sulfur density	g cm^{-3}	2.07
ρ_C	Carbon density	g cm^{-3}	2.26
Variables			
$load_S$	Sulfur loading	mg cm^{-2}	
$m_{C,rel}$	Carbon content	%	
por	Porosity of the cathode	%	
$C - rate$	C-rate		

investigated. Fig. 7(b) shows the simulation results of the expansion/contraction curves for four different sulfur loading. The relative sulfur content remains the same, but the absolute sulfur mass in the cathode will increase. This results in different geometric dimensions for the simulated cells, which are proportional to the sulfur loading. The capacity of the cells increases with rising sulfur loading, since more active material can be converted.

The parameter variation of porosity and carbon content shows similarities to each other. The two parameters have a major impact on the dimensions of the cell. In Fig. 7(c), the expansion curves are plotted against the summed capacitance for different porosities. The more porous the cathode, the thicker the individual layers. Thus, the initial thickness of the battery increases. In addition, more charge can be removed from or added to the cell. This effect is due to the change in the reactive sulfur content and the associated change in electrical conductivity. With the more porous structure, more conversion and nucleation sites are available.

Compared to the variation of porosity, the cathode thickness and thus the initial total thickness of the cell grows with increasing carbon content. By increasing the carbon content, the carbon matrix becomes larger, which raises the conductivity of the cathode. Thus, the migration of electrons from the current collector to the reaction sites of the insulating sulfur is enhanced. This effect increases the capacity, which can be removed from the cell. Compared to the variation of the sulfur loading, this effect is much smaller for the variation of the carbon content.

5.3. Validation of the simulations results

By studying the sensitivity of the selected parameters, it becomes observable how different the expansion curves behave depending to the

respective conditions. In order to validate the simulation results, the expansion behaviour is investigated on the one hand at the experimental expansion (see Fig. 6 (b)) and on the other hand by means of experimental investigations from [6]. Waluś et al. present the results of the volumetric changes of Oxis Energy's pouch cells under different operating conditions [6].

The experimental investigation of the volumetric change of the cell in Fig. 6(b) in relation to the simulated parameter variation in Fig. 7(a) shows the spline curve behavior. This is an indication of the individual conversion stages in the electrolyte. On the other hand, it becomes visible that the courses of the three successive charging steps become more linear. This can be related to the history of the cell. As the discharge rate increases, smaller Li_2S crystals form, which can be converted more quickly in the charging step. This results in a less curved thickness profile. It can also be seen that in Fig. 6(b) there is an irreversible overall change in thickness. This can be attributed to the added LiNO_3 in the electrolyte. During the reaction with the metallic lithium anode, irreversible gas formation can occur, so that if the cell is not strained, it expands. Such a cushion-shaped expansion of the pouch cell is observed in the measurements in Fig. 6(b). Due to the presence of the gas, internal structural damage can be a consequence, which could also show a deviation between simulation and measurement. Furthermore, the simulation shows an idealised reaction process. In the conversion process, it cannot be ruled out that different polysulfide chains could occur in parallel. In addition, Waluś et al. observed a transient process [6]. In our measurements, we cannot exclude whether a transient process is present or whether fast cyclic ageing of the pouch cell has occurred. Due to the prototype status, there can also be design-related irregularities in the manufacturing process. The test pouch cell is not uniformly thick across the cross-sectional area. The described aspects are potential explanations for the differences between the own measurement and the simulation. Due to these deviations, the simulation results are also compared with the measurements by Waluś et al. [6].

The change of thickness at the investigation at Waluś et al. can be identified up to 0.45mm [6]. Fig. 8 shows the two thickness profiles. Both thickness profiles are plotted versus the continuous capacity. Fig. 8 (a) illustrates the simulated thickness curve for the 5Ah test cell. Fig. 8 (b) shows the measured curve of Waluś et al. for an 11Ah Li-S pouch cell. This cell has dimensions of 155x102mm. The battery has been measured at a temperature of 30°C. Cycling has been performed between the voltage limits of 1.5–2.45V. The shown curve of Waluś et al. has been recorded at the 10th charge cycle of the cell [6]. The discharge is performed at a rate of 0.2 C and the charging at 0.1 C.

For the simulated thickness profile in Fig. 8(a), the resulting contraction during discharge is 0.326mm. For the reference battery in Fig. 8(b), a contraction during the discharge can be observed to be 0.345mm. Thus, both thickness changes are around the same order of magnitude. It can be seen that the thickness change is related to the SoC. The higher capacity, despite approximately the same thicknesses, can be explained by the larger cross-sectional area of the Oxis Energy cell. Thus, the cell from Oxis Energy has a larger volume with a similar content of sulfur. Thus, a larger amount of active material is also present. A further similarity between simulation and measurement is the curve shape. In both figures, a curved progression can be seen. The turning point for both profiles is also around the halfway of the discharge process. This confirms the assumed process according to the interpolated points. In the study of Waluś et al. during the charging process, the thickness change increases almost in a linear way [6]. This is not the case in the simulation. One reason could be the reduced voltage threshold, which leads to an earlier termination of the charge. As a result of an increased end-of-discharge voltage, the measurement results could deviate. In our own measurements, the curved progression is observed during charging (see Fig. 6). In addition, the differences may result from the unpublished composition of the battery of Oxis Energy, as well as from the assumed sequential processes in the simulation.

In order to describe the influence of different C-rates on the

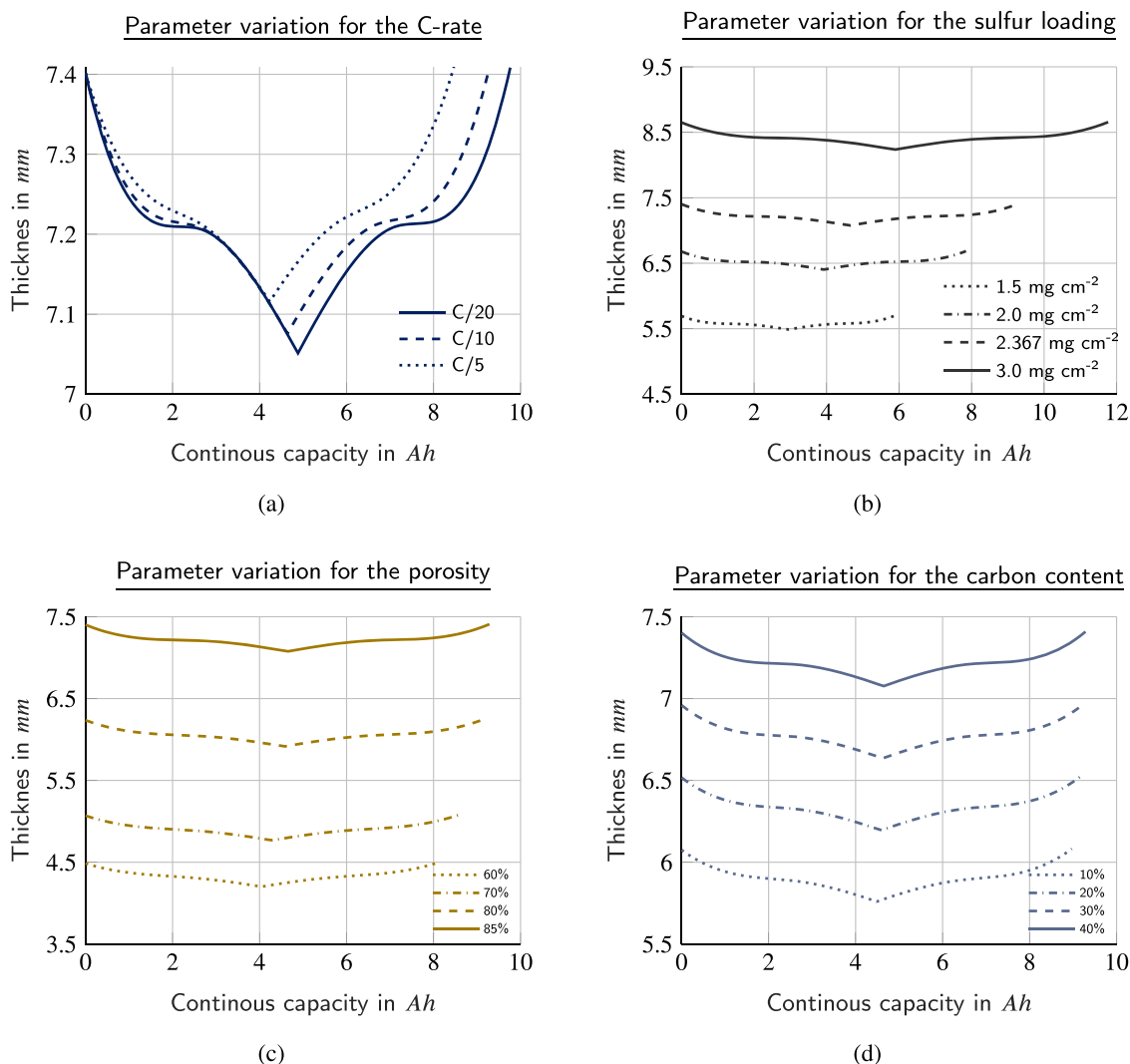


Fig. 7. Cell thickness and parameter variation of C-rates in (a), the different sulfur loading in (b), the different porosity of the cathode in (c), and the different carbon content on the cathode side in (d).

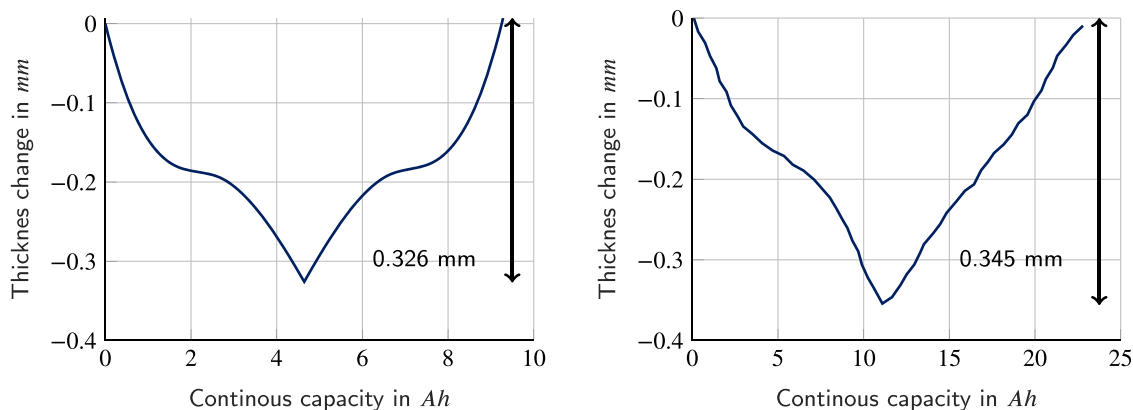


Fig. 8. Simulated cell thickness versus continuous capacity of the 5Ah test cell (a) and the measured cell thickness versus continuous capacity (b) of a 11Ah pouch cell from Oxis Energy by Waluś et al. [6].

volumetric changes of the cell, this effect has been investigated experimentally on the Li-S test cells. To compare the measurement from the literature with that in Section 4, it should be mentioned that Waluś et al. collected the experimental data with a laser measuring instrument [6]. Measurement data from 4 are recorded with a mechanical measuring tip,

so it is assumed that these measurement data are of greater uncertainty.

Waluś et al. show that the thickness evolution is mainly related to the expansion of the electrodes and not to the gas formation [6]. It is also shown in [6] that the volumetric expansion is mainly influenced by the metallic anode. The contraction, or expansion, of the cell in the

simulation is also mainly controlled by the reaction of the anode side. The characteristic spline shape is influenced by the processes at the cathode. Waluś et al. demonstrate this experimentally [6]. Different discharge currents result in different morphologies of Li_2S . This process is also considered in the model. It is further observable that higher C-rates result in a more stable thickness change (rather linear) (see Figs. 6 and 7). Waluś et al. also shows this observation in its experimental investigations [6]. The higher the C-rate selected, the lower the removed capacity and the more higher order polysulfides remain in the electrolyte.

The discharge is therefore not finally completed. Consequently, the recharging of the cell takes less time. This is also visible in the parameter variation of the C-rate in Fig. 7, as well as in the experimental investigations in Fig. 6. The aging is only basically taken into account by a mathematical term in the model. Waluś et al. show that determining the State of Health (SoH) by the change in thickness over time can be a useful parameter for a future BMS [6]. The fast aging of the cell is visible in the expansion curves in Fig. 6(b) in the experimental section, as in [6]. An offset between each cycle is evident. This can be attributed to the mossy growth of Li and the incomplete conversion of the solid polysulfides.

6. Conclusion

In this work, a mechanical model to describe the volumetric changes of a Li-S cell has been developed for different constant discharge and charge rates. The conversion process and the associated precipitation/dissolution of various polysulfide chains result in a volumetric change in the cell stack. The initiating effects of the volume change of the cell are described separately for the cathode and anode side. The shuttle mechanism is not considered in this model, because the electrolyte of the test cell is added with lithium nitrate, which inhibits this effect. Waluś et al. show that the volumetric expansion is not dominated by gas formation, but rather by the reactions within the cell [6]. In this case, the metallic lithium anode dominates the maximum contraction/expansion and masks the change of the volume of the cathode. The reactions taking place on the cathode side shape the typical spline curve. A linear behavior over the SoC is assumed for the reversible elongation of the lithium anode. With an increasing number of cycles, an irreversible expansion of the anode through dendrite formation is assumed. To model the cathode, four characteristic states of conversion of the active material and the associated changes in volume are defined.

Mathematical relationships are derived to describe the thickness change for the anode as well as for the cathode. The sum of the two thickness changes, aging effects and number of layers results in the total thickness change of the cell stack. The described model approach has been implemented and simulated in MATLAB/Simulink.

A expansion measurement at different C-rates is presented to verify the assumed model approach. The individual assumed conversion stages can be identified of the curve characteristics. For the parameterization of the model and the understanding of the structure of the cell, CT scans has been performed on the test cell. The scans show the size difference between anode and cathode, as well as the porous structure of the cathode. A variation of different parameters is performed to improve the interpretation of the sensitivity of these selected parameters on the simulation results. It is shown that even small adjustments can have a sizeable impact on the volumetric changes. Subsequently, the simulation results are discussed with an experimental study from [6] and own expansion measurements at different C-rates. The results of the simulation and the measurements show noticeable similarities quantitatively. These depend on the selected parameters of the model. It can be seen also that the expansion characteristic of the Li-S cell is related to the SoC. This could be shown by measurements as well as the simulation. Based on the results of the measurement and the simulation, this model approach could be considered in the development for a BMS of the Li-S battery.

CRedit authorship contribution statement

Daniel Martin Brieske: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Alexander Warnecke:** Conceptualization, Resources, Data curation, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Dirk Uwe Sauer:** Conceptualization, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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